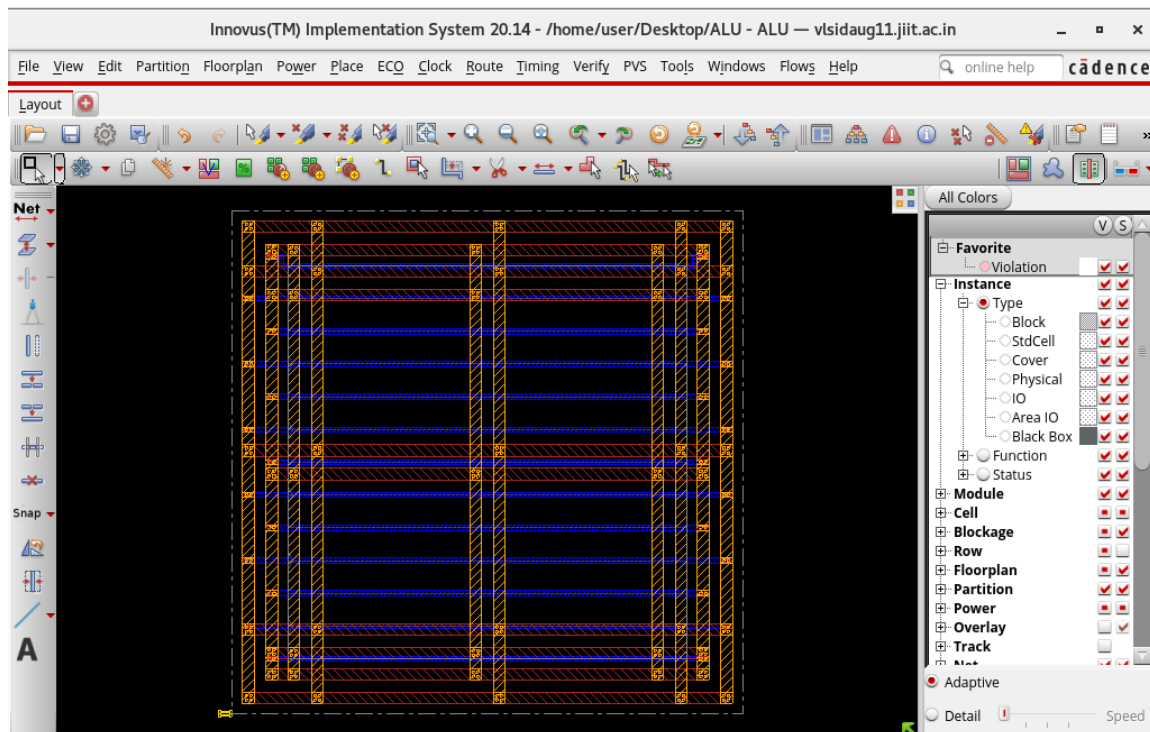


DESIGN PROCESS STAGES

1. SPECIAL ROUTED ALU



2. PLACED (UNOPTIMISED)

Applications Places Terminal

user@vlsidaug11:ALU

File Edit View Search Terminal Help

Total Power

Total Internal Power:	0.00893475	56.4359%
Total Switching Power:	0.00688347	43.4784%
Total Leakage Power:	0.00001372	0.0867%
Total Power:	0.01583195	

Group	Internal Power	Switching Power	Leakage Power	Total Power	Percentage (%)
Sequential	0	0	0	0	0
Macro	0	0	0	0	0
IO	0	0	0	0	0
Combinational	0.008935	0.006883	1.372e-05	0.01583	100
Clock (Combinational)	0	0	0	0	0
Clock (Sequential)	0	0	0	0	0
Total	0.008935	0.006883	1.372e-05	0.01583	100

Rail	Voltage	Internal Power	Switching Power	Leakage Power	Total Power	Percentage (%)
VDD	0.9	0.008935	0.006883	1.372e-05	0.01583	100

* Power Distribution Summary:

* Highest Average Power: g4576 (ADDFX4): 0.0003705

* Highest Leakage Power: g4576 (ADDFX4): 5.141e-07

* Total Cap: 7.17957e-13 F

* Total instances in design: 174

* Total instances in design with no power: 0

* Total instances in design with no activity: 0

* Total Fillers and Decap: 0

Applications Places Terminal

user@vlsidaug11:ALU

File Edit View Search Terminal Help

timeDesign Summary

Setup views included:
WORST

Setup mode	all	default
WNS (ns):	0.000	0.000
TNS (ns):	0.000	0.000
Violating Paths:	0	0
All Paths:	0	0

DRVs	Real	Total
Nr nets (terms)	Worst Vio	Nr nets (terms)
max_cap	0 (0)	0 (0)
max_tran	0 (0)	0 (0)
max_fanout	0 (0)	0 (0)
max_length	0 (0)	0 (0)

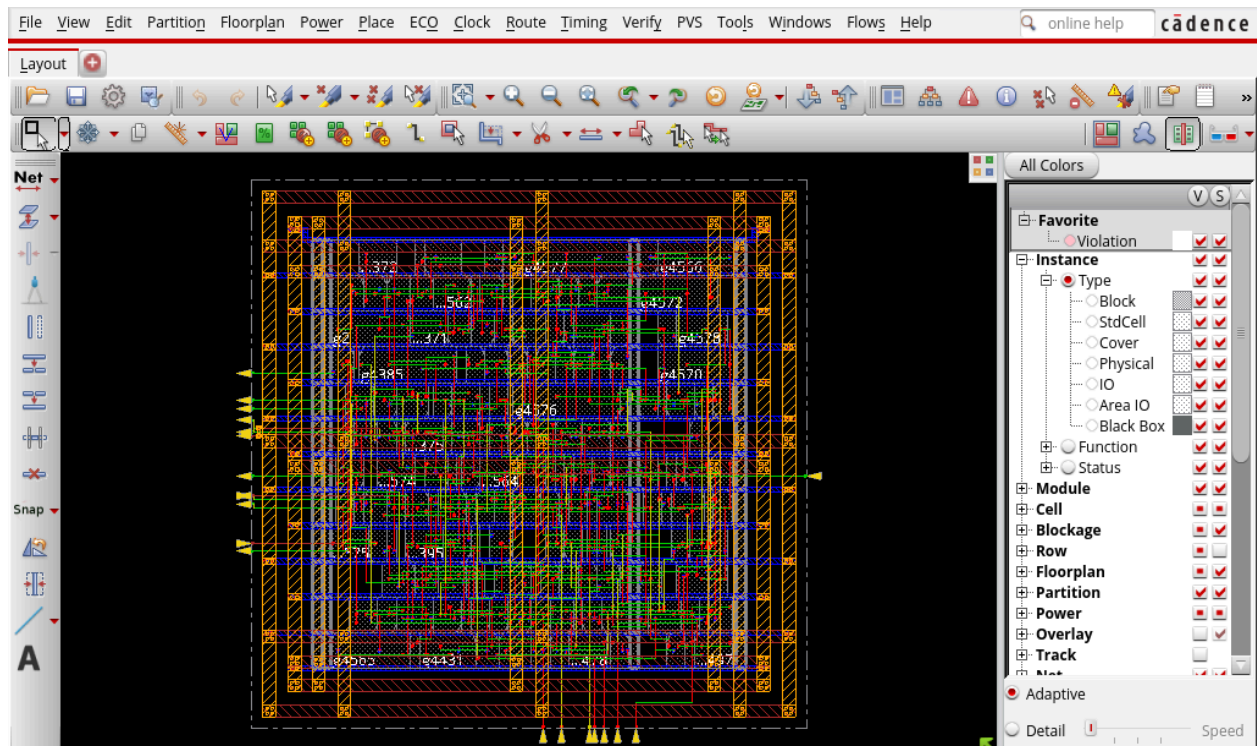
Density: 87.353%
Routing Overflow: 0.00% H and 0.00% V

Reported timing to dir timingReports
Total CPU time: 0.14 sec
Total Real time: 0.0 sec
Total Memory Usage: 1486.457031 Mbytes
*** timeDesign #1 [finish] : cpu/real = 0:00:00.1/0:00:00.9 (0.2), totSession cpu/real = 0:00:42.1/0:06:31

innovus 1> report_area

Inst Name	Module Name	Inst Count	Total Area
ALU		174	2963.822

innovus 2> report_power



3. PLACED (OPTIMISED)

Applications Places Terminal

user@vlsidaug11:ALU

File Edit View Search Terminal Help

Total Power

Total Internal Power:	0.00679306	56.0831%
Total Switching Power:	0.00676002	49.8395%
Total Leakage Power:	0.00081049	0.0773%
Total Power:	0.01356357	

Group	Internal Power	Switching Power	Leakage Power	Total Power	Percentage (%)
Sequential	0	0	0	0	0
Macro	0	0	0	0	0
I/O	0	0	0	0	0
Combinational	0.006793	0.00676	1.049e-05	0.01356	100
Clock (Combinational)	0	0	0	0	0
Clock (Sequential)	0	0	0	0	0
Total	0.006793	0.00676	1.049e-05	0.01356	100

Rail	Voltage	Internal Power	Switching Power	Leakage Power	Total Power	Percentage (%)
VDD	0.9	0.006793	0.00676	1.049e-05	0.01356	100

Power Distribution Summary:

Highest Average Power:

g4576 (ADDFX2): 0.0002842

Highest Leakage Power:

g4576 (ADDFX2): 2.626e-07

Total Cap:

7.13776e-13 F

Total instances in design: 174

Total instances in design with no power: 0

Total instances in design with no activity: 0

Total Fillers and Decap: 0

Applications Places Terminal

user@vlsidaug11:ALU

File Edit View Search Terminal Help

Setup views included:

WORST

Setup mode	all	default
WNS (ns):	0.000	0.000
TNS (ns):	0.000	0.000
Violating Paths:	0	0
All Paths:	0	0

DRVs	Real	Total
	Nr nets (terms)	Worst Vio
max_cap	0 (0)	0 (0)
max_tran	0 (0)	0.000
max_fanout	0 (0)	0 (0)
max_length	0 (0)	0 (0)

Density: 78.333%

Routing Overflow: 0.00% H and 0.00% V

Reported timing to dir timingReports

Total CPU time: 0.06 sec

Total Real time: 1.0 sec

Total Memory Usage: 1523.71875 Mbytes

*** timeDesign #2 [finish] : cpu/real = 0:00:00.1/0:00:01.0 (0.1), totSession cpu/rea

innovus 3> report_area

Inst Name	Module Name	Inst Count	Total Area
ALU		174	2657.794

innovus 4> report_power

Using Power View: WORST.

Set Default Frequency 100MHz.

[user@vlsidaug11:ALU] ALU user@vlsidaug11:ALU Innovus(TM) I

[user@vlsidaug11:ALU] ALU user@vlsidaug11:ALU

File View Edit Partition Floorplan Power Place ECO Clock Route Timing Verify PVS Tools Windows Flows Help

online help cadence

Layout

Net

Snap

A

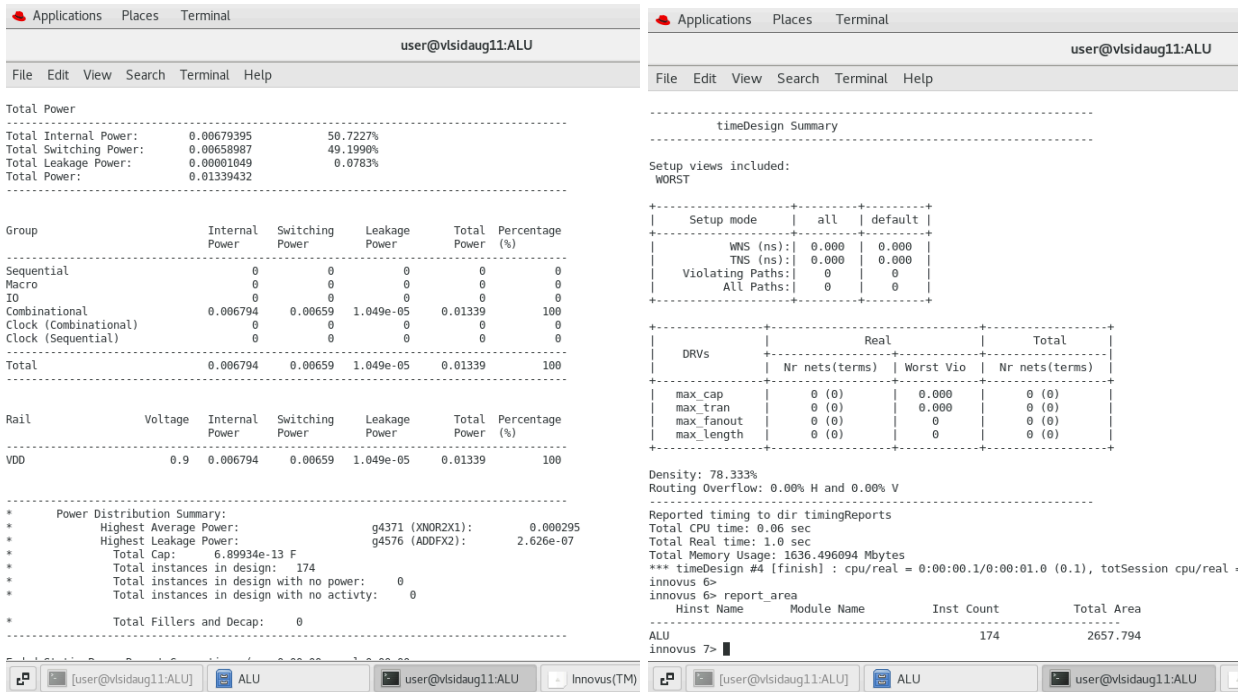
select Instance Enter Instance name.

Append coords 0,0 radius 150 Goto

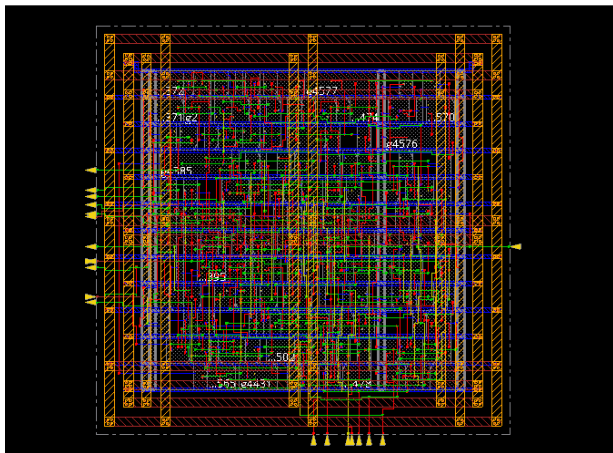
79.95950, 76.47900 Sel: 0 Timing Analyzed

4. Nanorouted pre and post optimisation -

Reports post



Pre



Post

